

Alexandria University
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HVDC Transmission System

12-Pulse LCC-HVDC System: Control and Fault Analysis

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Simulink Model Link:
Click here to access the Simulink Model on GitHub

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Abstract

This report presents the modeling and simulation of a 1.56 GW, 200 kV 12-pulse Line-Commutated Converter (LCC) HVDC transmission system. The study evaluates the system's dynamic performance under normal operating conditions utilizing Constant Current (CC) and Constant Extinction Angle (CEA) control strategies. Furthermore, the system's resilience is tested under a severe 25% AC voltage sag at the rectifier side, demonstrating a successful automated control handover mechanism to Constant Ignition Angle (CIA) and Inverter CC mode.

Contents

1 System Specifications **3**

2 Case A: Normal Operating Conditions **3**
2.1 System Configuration (Case A) 3
2.2 Overall System Performance (Case A) 5
2.3 Detailed Waveform Analysis 6

3 Case B: Severe Sag Condition **9**
3.1 System Configuration (Case B Modifications) 9
3.2 Overall System Performance (Case B) 11
3.3 Detailed Waveform Analysis 12

4 Conclusion **14**

1 System Specifications

The system parameters utilized for the simulation are derived from the project requirements and are detailed in Table 1.

Table 1: HVDC System Simulation Parameters

Parameter	**Value**
Rated Power	1.56 GW
DC Link Voltage	+200 kV
DC Line Resistance	5 Ω
Bus A (Rectifier) AC Voltage	75.19 kV (rms)
Bus B (Inverter) AC Voltage	77.618 kV (rms)
System Frequency	50 Hz

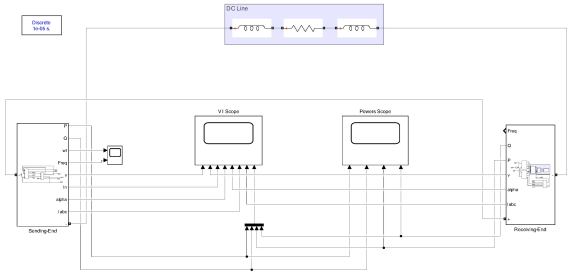
2 Case A: Normal Operating Conditions

In this scenario, the HVDC transmission system operates under steady and balanced grid conditions, aiming to maintain a nominal target DC voltage of 200 kV. The coordination between the two converter stations uses a dual-terminal control strategy of the kind commonly used for LCC-HVDC links. The Rectifier station acts as the primary current regulator, employing a Constant Current (CC) control loop. It continuously adjusts its firing angle (α) to minimize the error between the measured DC link current and the dynamic reference signal.

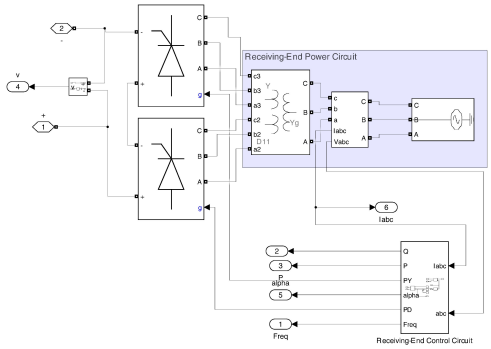
Conversely, the Inverter station operates under Constant Extinction Angle (CEA) control. It maintains the extinction angle (γ) at a safe predetermined value (typically 40°) to establish the DC voltage profile and provide a sufficient margin to prevent commutation failures. Furthermore, this case evaluates the system's dynamic tracking capability by applying sudden step changes to the current reference, observing the response of the PI controllers.

2.1 System Configuration (Case A)

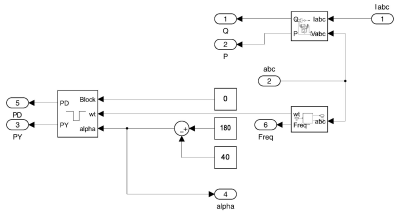
Figure 1 illustrates the internal subsystems utilized during normal operation, detailing the original wiring and control logic before any grid faults are introduced. The configuration prominently features 12-pulse thyristor converters at both ends to significantly mitigate 5th and 7th AC harmonic distortions. The control circuits are equipped with Proportional-Integral (PI) controllers meticulously tuned to deliver a fast dynamic response with minimal overshoot.



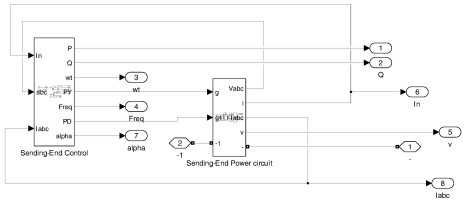
(a) **Main System:** Overall architecture.



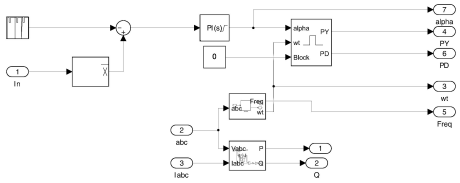
(b) **Receiving Power Circuit:** Transformer and bridges.



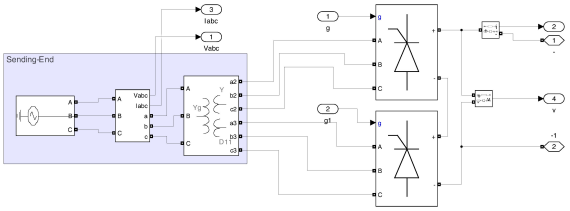
(c) **Receiving Control Circuit:** Original CEA logic.



(d) **Sending Control Circuit:** CC control logic.



(e) **Pulse Generator Unit:** Synchronization block.



(f) **Sending Power Circuit:** Rectification bridges.

Figure 1: Subsystem Configurations for Case A (Normal Operation).

2.2 Overall System Performance (Case A)

Figures 2 and 3 display the combined scopes of the system's dynamic response to step changes in the current reference.

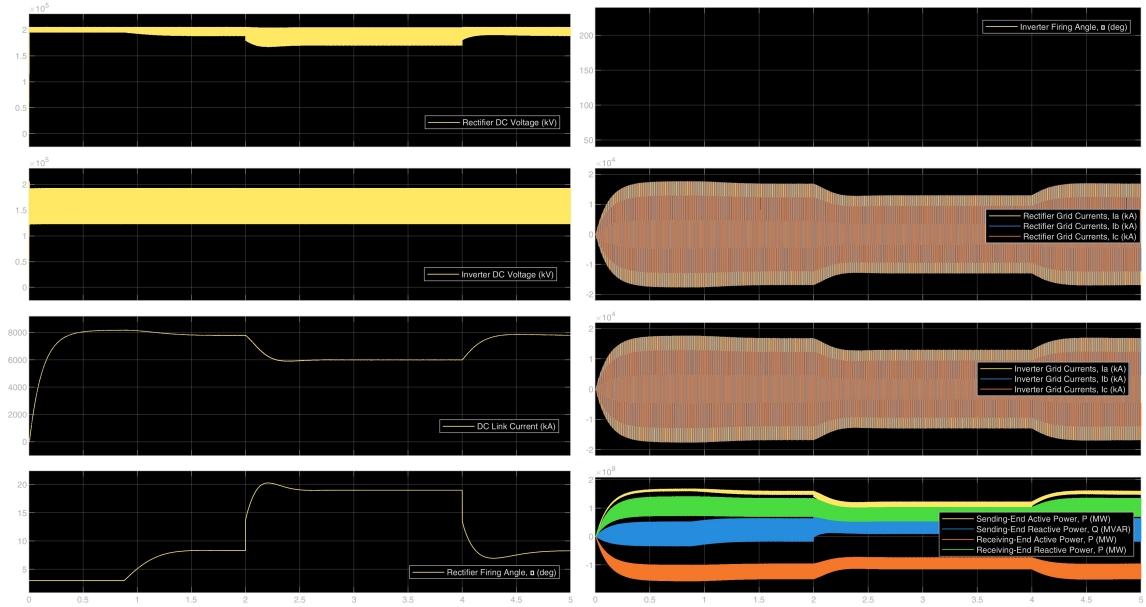


Figure 2: **VI Scope:** Displays the synchronization between voltages, currents, and firing angles as the system accurately tracks the step references under normal grid conditions.

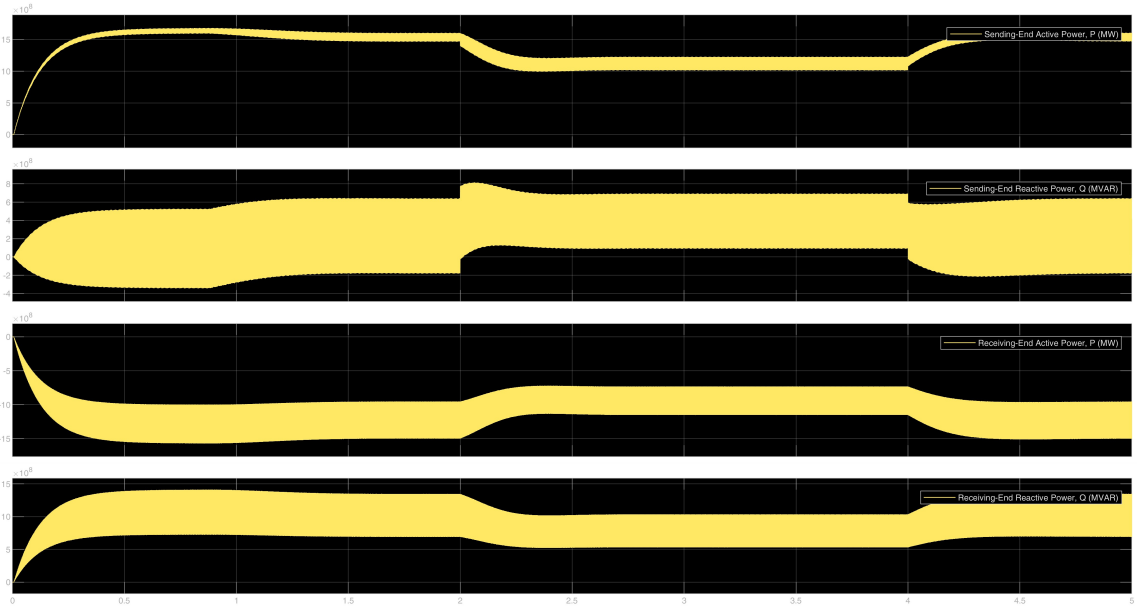


Figure 3: **Powers Scope:** Illustrates the corresponding active power (P) transmission and reactive power (Q) consumption profiles at both ends.

2.3 Detailed Waveform Analysis

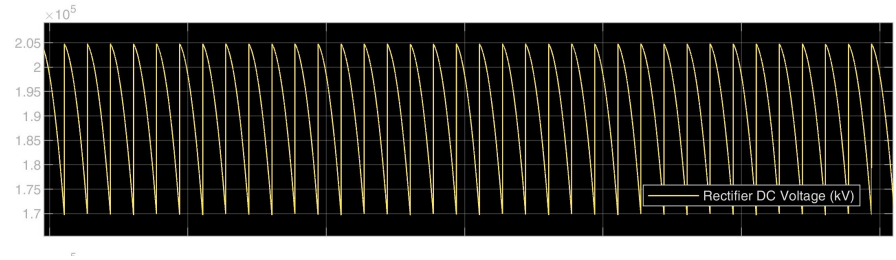


Figure 4: **Rectifier DC Voltage (V_{rect})**

The sending-end DC voltage remains stable around the nominal 200 kV mark. The waveform clearly exhibits the characteristic 12-pulse ripples (12 peaks per AC cycle). The 12-pulse configuration gives less ripple than a 6-pulse bridge, which lowers the size requirement for the DC smoothing reactor.

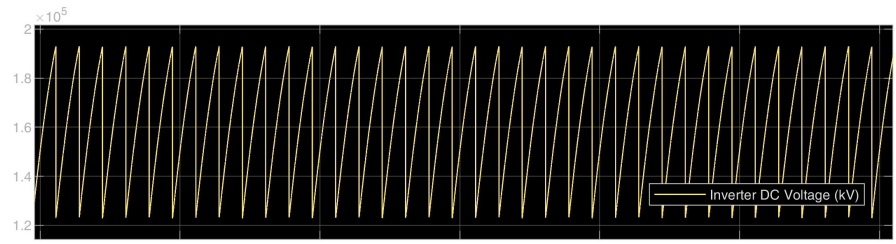


Figure 5: **Inverter DC Voltage (V_{inv})**

The receiving-end DC voltage acts as a controlled back-EMF. Operating in the inversion region ($\alpha > 90°$), it maintains a negative average voltage. The potential difference (ΔV) between the rectifier and the inverter is the driving force that pushes the DC link current across the transmission line resistance.

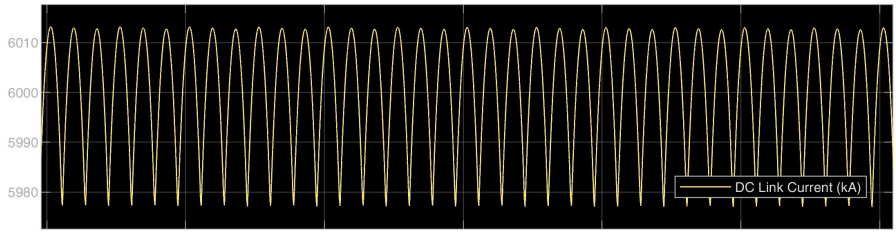


Figure 6: **DC Link Current (I_{dc})**

The DC current follows the applied reference signal. Upon the application of step changes (e.g., at $t = 2s$ and $t = 4s$), the current transitions smoothly with a short settling time, no visible steady-state error, and small overshoot. This indicates that the rectifier's PI current controller is well tuned.

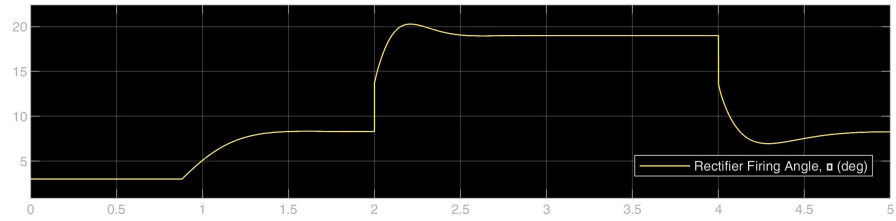


Figure 7: **Rectifier Firing Angle (α_{rect})**

The rectifier firing angle dynamically modulates within the active rectification zone (typically 10° to 30° during steady state) to strictly regulate the DC current. When a lower current is demanded, the PI controller seamlessly increases the firing angle to lower the output voltage, thereby reducing the line current.

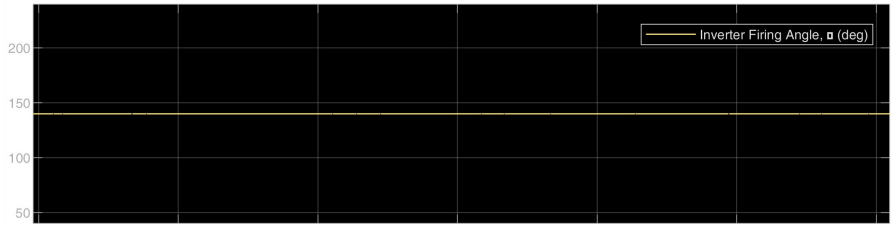


Figure 8: **Inverter Firing Angle (γ_{inv})**

The inverter operates under Constant Extinction Angle (CEA) control, firmly maintaining γ at 40°. This fixed margin ensures that the thyristors have sufficient time to regain their forward-blocking capability before the voltage polarity reverses, effectively protecting the inverter against catastrophic commutation failures.

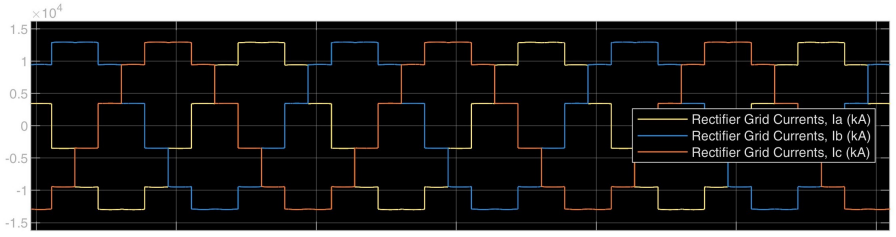


Figure 9: **Rectifier AC Grid Currents**

The three-phase AC currents drawn from the sending-end grid exhibit a balanced quasi-square waveform. The implementation of the $Y-Y$ and $Y-\Delta$ transformer arrangements inherently cancels out the 5th and 7th harmonics, resulting in a cleaner 12-step waveform that approximates a sinusoidal shape, reducing filtering requirements.

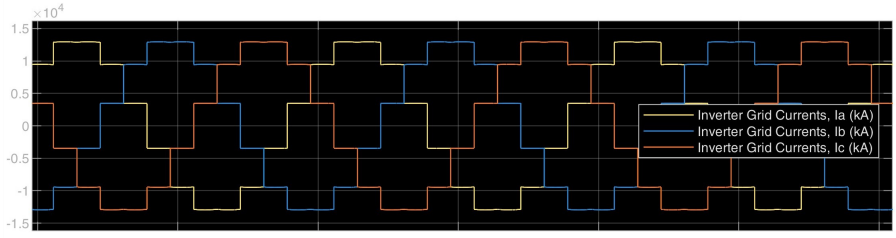


Figure 10: **Inverter AC Grid Currents**

Similar to the rectifier, the AC currents injected into the receiving grid are balanced and low in harmonic distortion. The amplitude of these currents is directly proportional to the active power being delivered to the receiving network.

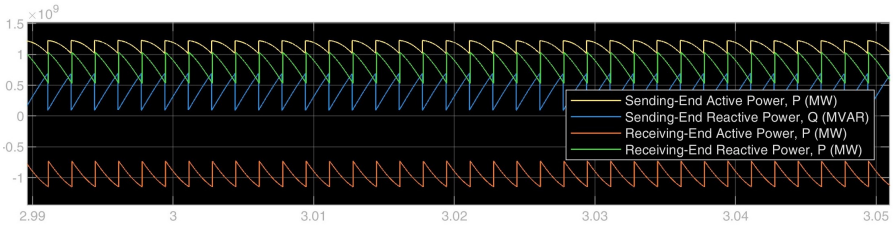


Figure 11: **Active and Reactive Power (P&Q)**

The active power (P) accurately tracks the step changes of the DC current, proving efficient energy transfer. Meanwhile, Line-Commutated Converters (LCCs) inevitably consume reactive power (Q). The reactive power demand is clearly visible and fluctuates directly with the firing angles; larger firing angles result in higher reactive power consumption from the AC grid.

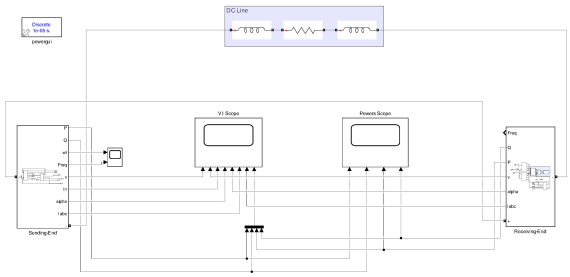
3 Case B: Severe Sag Condition

This case investigates the system's dynamic resilience and control adaptability during a severe 25% voltage sag occurring at the rectifier's AC grid. When the sending-end AC voltage drops significantly, the rectifier struggles to generate sufficient DC voltage to drive the commanded DC current across the transmission line. In an attempt to compensate, the rectifier's PI controller aggressively decreases the firing angle until it hits the absolute hardware lower limit ($\alpha_{min} = 3^\circ$).

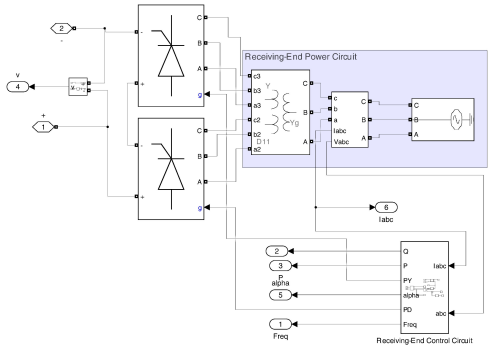
At this saturation point, the rectifier falls into Constant Ignition Angle (CIA) mode and entirely loses its ability to regulate the current. To prevent the DC link current from collapsing to zero, a contingency control strategy dictates that the Inverter must seamlessly take over the Constant Current (CC) regulation. This operational transition is achieved by subtracting a 5% Current Margin from the original current reference, ensuring that the inverter's CC controller becomes active and assumes total control of the power flow to sustain grid stability.

3.1 System Configuration (Case B Modifications)

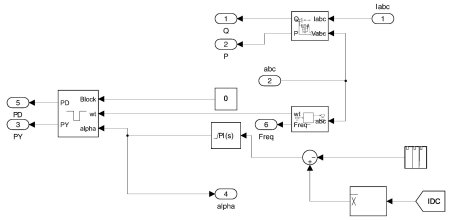
Figure 12 details the system configuration after implementing the necessary modifications to facilitate the current control handover for Case B. A `Goto` global tag (labeled `IDC`) was attached to the sending-end DC current measurement block. This signal is wirelessly routed to a corresponding `From` tag inside the receiving-end control subsystem, allowing the inverter to modulate its firing angle dynamically and effectively replace the rectifier as the primary current regulator.



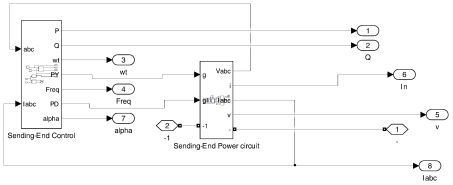
(a) **Main System:**



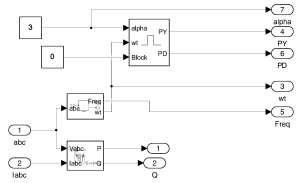
(b) **Receiving Power Circuit:**



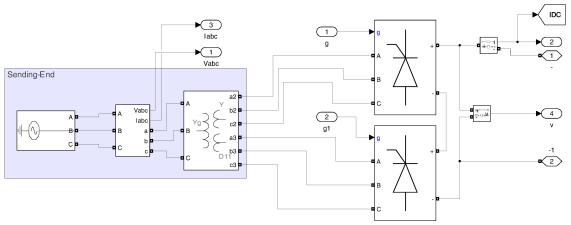
(c) **Receiving Control Circuit:** Includes IDC measurement.



(d) **Sending Control Circuit:**



(e) **Pulse Generator Unit:**



(f) **Sending Power Circuit:** Emitting the IDC signal.

Figure 12: Subsystem Configurations modified for Case B.

3.2 Overall System Performance (Case B)

Figures 13 and 14 demonstrate the successful control handover during the 25% voltage sag.

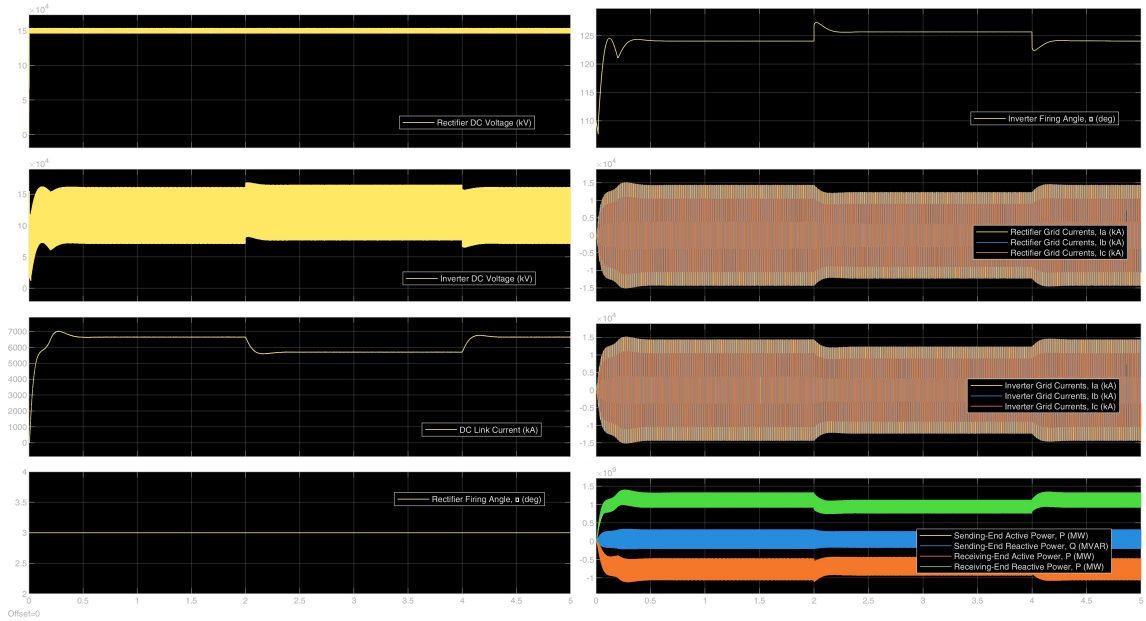


Figure 13: **VI Scope:** Highlights the critical transition point where the inverter abandons CEA mode and successfully takes over the Constant Current (CC) regulation.

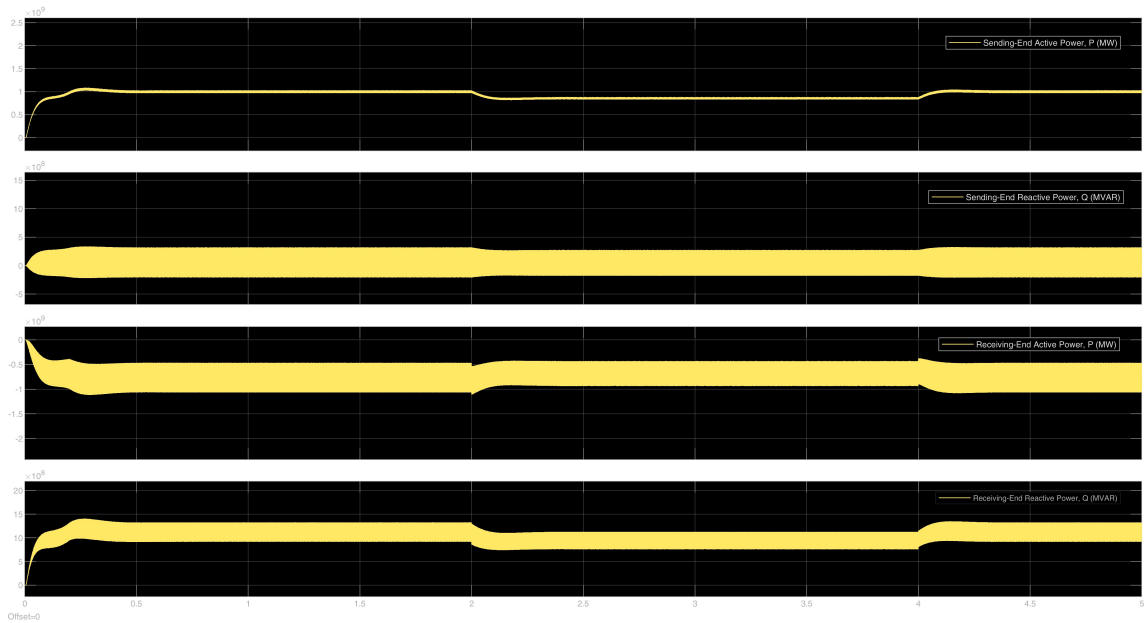


Figure 14: **Powers Scope:** Shows the resulting active and reactive power transfer limitations dictated by the severe voltage sag on the sending end.

3.3 Detailed Waveform Analysis

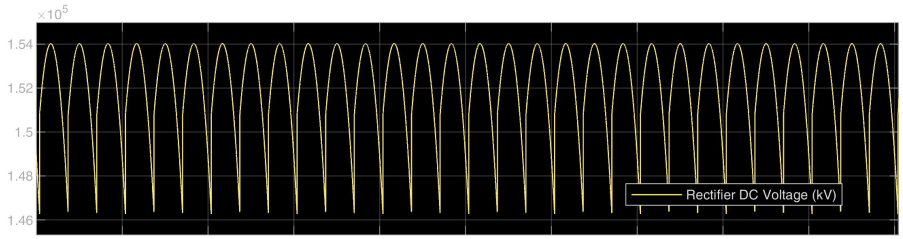


Figure 15: **Rectifier DC Voltage** (V_{rect})

A clear and sustained drop in the DC voltage (falling to approximately 150 kV) is observed. This is the direct physical consequence of the 25% AC voltage sag applied to the sending-end source, fundamentally reducing the maximum achievable DC output capability of the rectifier bridge.

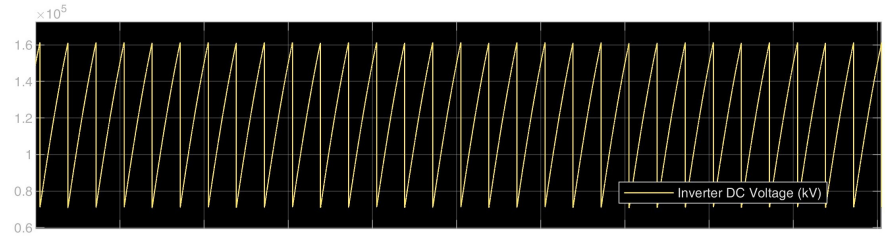


Figure 16: **Inverter DC Voltage** (V_{inv})

To maintain the flow of current against the collapsed rectifier voltage, the inverter dynamically reduces the magnitude of its opposing back-EMF. By adjusting its voltage downwards, the inverter guarantees that the forward potential difference (ΔV) remains positive and sufficient to drive the margin-adjusted current.

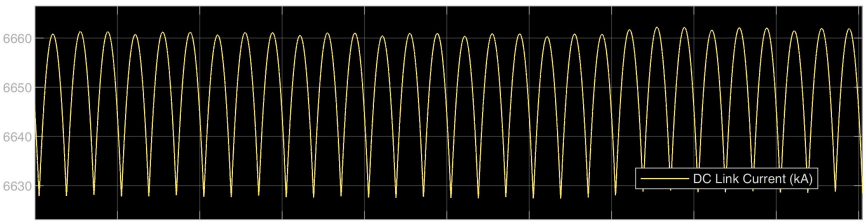


Figure 17: **DC Link Current** (I_{dc})

Despite the severe fault, the DC current does not collapse. It smoothly transitions to track the new, lower reference points established by the 5% current margin (e.g., adjusting from 7000 A down to 6650 A). This waveform shows that the inverter's CC controller has assumed command of the link.

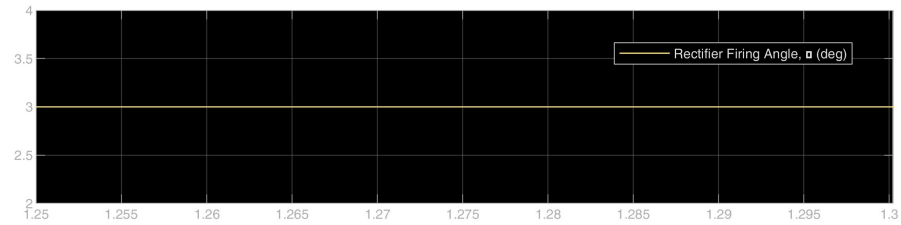


Figure 18: **Rectifier Firing Angle** (α_{rect})

The rectifier's PI controller integrates to its absolute minimum in a futile attempt to raise the voltage, causing the firing angle to saturate and flatline at $\alpha_{min} = 3°$. The rectifier is now effectively operating as an uncontrolled diode bridge (Constant Ignition Angle - CIA mode) to squeeze out the maximum possible voltage.

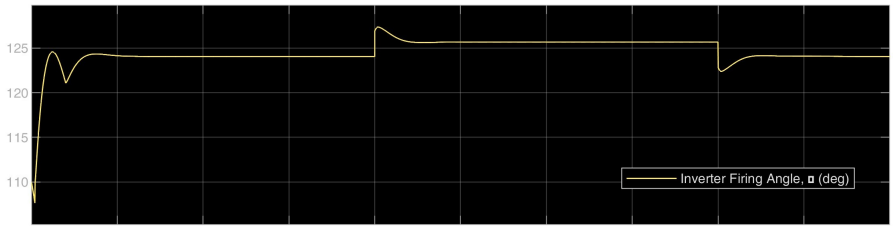


Figure 19: **Inverter Firing Angle** (α_{inv})

In stark contrast to Case A, the inverter firing angle is no longer fixed at a safe margin. The inverter abandons CEA mode and actively modulates α_{inv} to regulate the DC line current, proving the operational execution of the fallback CC control loop.

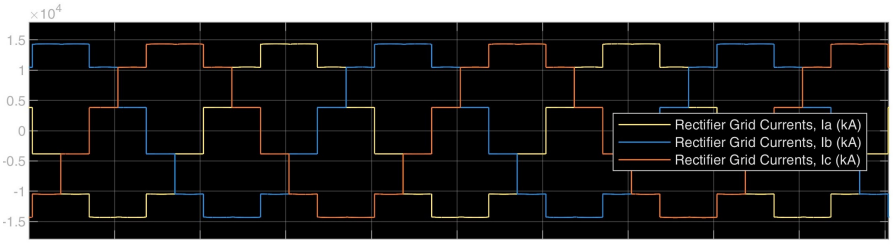


Figure 20: **Rectifier Grid Currents**

While the quasi-square shape of the currents remains intact, their overall magnitudes are visibly reduced. This reduction corresponds to the newly enforced 5% margin drop in the DC link current caused by the control handover.

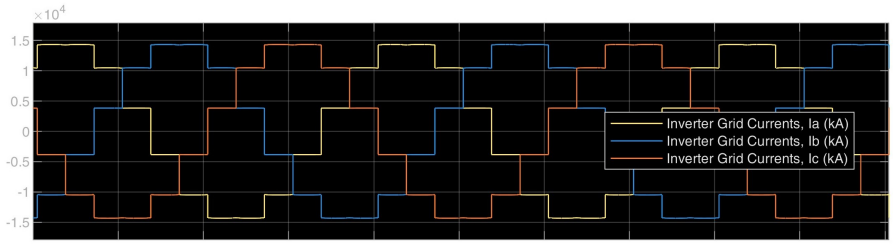


Figure 21: **Inverter Grid Currents**

The AC currents injected into the receiving grid accurately mirror the new current margins. The ability of the inverter to continue supplying stable, balanced, and low-harmonic AC currents during a severe sending-end fault shows that the link rides through this fault in the simulation.

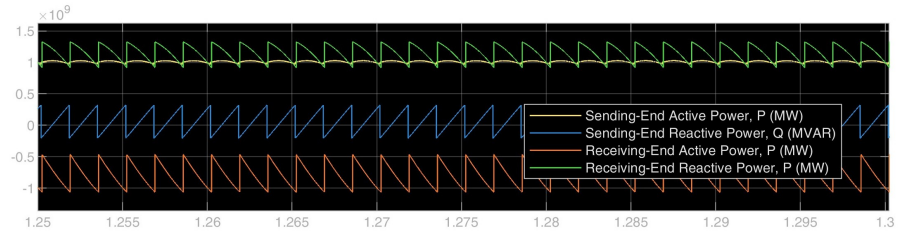


Figure 22: **Active and Reactive Power (P&Q)**

Due to the significant loss of DC voltage, the overall active power (P) transfer capacity is inherently reduced. Simultaneously, the reactive power (Q) demand shifts drastically due to the new extreme firing angles, highlighting the profound impact grid faults have on converter efficiency and MVAR requirements.

4 Conclusion

The comprehensive simulation of the 12-pulse HVDC transmission system shows its steady-state accuracy and its dynamic behavior. Under normal conditions, the primary CC/CEA control strategy ensures fast, accurate, and stable power transfer. Crucially, during the severe AC voltage sag, the system kept operating. The coordinated control handover—where the rectifier saturated into CIA mode and the inverter seamlessly adopted CC mode using a 5% current margin—prevented the DC-link current from collapsing. The results show the importance of coordinated control logic in maintaining power continuity under AC voltage disturbances.